

EXERCISE 4

BARBARA FENWU



EX 8.2.1

SOURCE	DESTINATION			SUPPLY
	1	2	3	
1	15	9	13	7
2	11	12	17	5
3	9	11	9	3
DEMAND	7	3	5	

USE the NORTHWEST CORNER rule to construct
A COMPLETE BFS

EX 8.2.2

	1	2	3	4	5	SUPPLY
1	2	4	6	5	7	4
2	7	6	3	4	4	6
3	8	7	5	2	5	6
4	0	0	0	0	0	4
DEMAND	4	4	2	5	5	

FIND AN INITIAL BFS WITH THE NORTHWEST CORNER RULE

EX 8.2.6

	1	2	3	4	5	supply
1	8	6	3	7	5	20
2	5	14	8	4	7	30
3	6	3	9	6	8	30
$u(D)$	0	0	0	0	0	20
DEMAND	25	25	20	10	20	

AFTER SOME ITERATIONS, THE BFS IS

$$\begin{array}{lll}
 x_{13} = 20 & x_{32} = 25 & x_{43} = 0 \\
 x_{21} = 25 & x_{34} = 5 & x_{45} = 20 \\
 x_{24} = 5 & x_{42} = 0 &
 \end{array}$$

STARTING HERE, DO TWO ITERATIONS OF THE TRANSPORTATION SIMPLEX METHOD

(HINT : $u_2 = 0$)

OTHER EXERCISES YOU CAN DO:

- 8.2.7(a)

- 8.2.8

- 8.2.17